

Hardik Sharma

PhD Researcher — Computer Vision — Deepfake Detection — Multimodal AI

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RESEARCH PROFILE

Computer vision researcher specializing in deepfake detection, multimodal document forensics, and explainable forensic reasoning. Research interests span self-supervised and reinforcement learning-based approaches to deepfake detection, chain-of-thought reasoning for evidence-grounded forensic analysis, and agent-based detection frameworks for real-world media authentication. Experienced in designing robust visual recognition systems that operate effectively under real-world degradations such as compression, noise, and distribution shifts.

Hands-on experience spans both research and production environments, including the development of scalable KYC verification systems and forensic pipelines leveraging ensemble methods and multimodal cues. Skilled in bridging theory and practice, with a particular interest in building reliable, interpretable, and generalizable vision systems for high-stakes applications such as digital identity verification and media authenticity.

EDUCATION

Indian Institute of Technology, Mandi

Ph.D. in Artificial Intelligence and Robotics, Centre for AI and Robotics

Indian Institute of Technology, Madras

B.S. in Data Science and Applications CGPA: 7.93

Partap World School

Class XII (CBSE) 88.2%

Partap World School

Class X (CBSE) 88.4%

Mandi, Himachal Pradesh

Aug 2025 – Present

Chennai, Tamil Nadu

Apr 2021 – May 2025

Indora, Himachal Pradesh

Apr 2019 – Mar 2020

Indora, Himachal Pradesh

Apr 2017 - Mar 2018

RESEARCH INTERESTS

- Computer Vision, Deepfake Detection, Multimodal Forensics, Explainable AI
- Chain-of-Thought Reasoning for Forensic Analysis; Actionable Deepfake Detection
- Reinforcement Learning and Self-Supervised Methods for Deepfake Detection
- Agent-Based Detection Frameworks; Tool-Augmented Forensic Pipelines
- AI-Generated Image Detection, Representation Learning, Robustness under Real-World Degradations
- Document Forensics, Visual Authentication, Spectral Learning

TECHNICAL SKILLS

- **Core Areas:** Computer Vision, Deep Learning, Multimodal Learning, NLP
- **Frameworks/Libraries:** PyTorch, TensorFlow, OpenCV, Scikit-learn, Hugging Face, Keras, NLTK, spaCy
- **Tools:** Python, SQL, Java, Git, Docker, Flask, FastAPI, LangChain, n8n

SELECTED PUBLICATIONS

- **CIRSNet: A Lightweight Chemometric Infrared Spectral Network for Spectral Analysis.** IEEE Transactions on Instrumentation and Measurement, 2025. [Link]
- **BioFASTNet: Biomedical Fragmented Attention Spectral Transformer Network for Infrared Spectra Analysis.** IEEE Transactions on Instrumentation and Measurement, 2026. [Link]
- **TRIDENT: Robust Deepfake Detection via Tri-Modal Forensic Ensembles.** In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2026. Accepted.
- **Beyond Backbones: Degradation-Aware Prototype Fusion for Robust Deepfake Detection.** In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2026. Accepted.
- **Deepfake Image Detection in the Wild: NTIRE 2026 Challenge Report.** CVPR Workshops, 2026. [Link]
- **Robust Deepfake Detection, NTIRE 2026 Challenge: Report.** CVPR Workshops, 2026.
- **Low Light Image Enhancement Challenge at NTIRE 2026.** CVPR Workshops, 2026.
- **DeepID Challenge of Detecting Synthetic Manipulations in ID Documents.** In ICCV 2025 Workshops, 2025. [Link]
- **ForensicCoT: Evidence-Augmented Chain-of-Thought Reasoning for Cross-Domain Text Manipulation Detection.** Submitted to ACM Multimedia, 2026.
- **DFToolBench-A: A Benchmark for Tool-Augmented Audio Deepfake Detection.** Submitted to Interspeech, 2026.
- **Text Summarization of Publicly Available BBC News Dataset on Kaggle.** Global Journal for Research Analysis (GJRA), Feb 2024. [Link]

RESEARCH EXPERIENCE

Vision Engineer

Apr 2024 – Jul 2025

Reagvis Labs

Remote

- Worked on visual forensic pipelines for fraud detection by integrating VLM-based cues with forensic priors and the TruFor algorithm.
- Improved challenge performance by combining baseline outputs with statistical measures, increasing weighted F1 from 0.706 to 0.712 and achieving 0.839 on FantasyID.
- Contributed to challenge-facing deepfake and document-forensics systems under realistic evaluation settings.

Computer Vision Intern

Dec 2024 – Apr 2025

Truens AI

Remote

- Developed a real-time deepfake detection system for banking KYC video calls, achieving **73.8% F1** on a private dataset and **70% aggregate F1** across multiple evaluation scenarios.
- Engineered a multimodal ID-card verification system combining computer vision and NLP to detect both visual and textual forgeries in identity documents.
- Focused on practical deployment constraints for remote onboarding and fraud-prevention workflows.

Research Intern

May 2024 – Present

Indian Institute of Technology, Mandi

Himachal Pradesh

- Conducted research on infrared spectral learning and chemometric modeling for molecular analysis.
- Built deep architectures for imbalanced multi-label prediction from spectral inputs, improving precision to **90.4%** and specificity to **97.9%** on odor prediction tasks.
- Applied transformer-based modeling and augmentation strategies on a dataset of **3,036 molecules** with **109 odor perception labels**.

Data Science Intern

May 2024 – Nov 2024

DeepAlgorithms Pvt. Ltd.

Remote

- Developed a behavioral biometric authentication system using motion-sensor data, achieving **88%** authentication accuracy with **FAR: 0.04** and **FRR: 0.08**.
- Engineered a spatial-attention-based deep model that processed raw accelerometer, gyroscope, and magnetometer inputs without manual feature engineering.

Research Intern

Jul 2024 – Dec 2024

AIISC, South Carolina

Remote

- Implemented jailbreak techniques from the Trees of Attack framework and evaluated vulnerability across **22** major LLMs.
- Benchmarked attack success across models including GPT-4, Claude, and LLaMA; while this is outside CV, it demonstrates rigorous empirical research and evaluation capability.

COMPETITIONS & CHALLENGE REPORTS

- **DeepID 2025 Challenge – Detecting Synthetic Manipulations in ID Documents** (ICCV 2025 Workshops): Ranked **5th out of 250+ teams**; focused on forged ID detection and localization under realistic KYC evaluation settings.
- **NTIRE 2026 Challenge – Robust AI-Generated Image Detection in the Wild** (CVPR 2026 Workshops): Ranked **5th out of 300+ teams**; contributed to robust detection of AI-generated images under diverse real-world transformations.
- **NTIRE 2026 Challenge – Robust Deepfake Detection** (CVPR 2026 Workshops): Ranked **6th out of 100+ teams**; developed forensic ensemble approaches for detecting deepfakes under challenging conditions.
- **Low Light Image Enhancement Challenge at NTIRE 2026** (CVPR 2026 Workshops): Ranked **9th out of 150+ teams**; contributed to image quality restoration in extreme low-light conditions.

SELECTED RESEARCH HIGHLIGHTS

- Published **CIRSNet** in *IEEE Transactions on Instrumentation and Measurement* (2025) — a lightweight chemometric network for infrared spectral analysis targeting chemical group identification and bio-magnetism; deployed on **ESP32** for embedded real-time inference.
- Published **BioFASTNet** in *IEEE Transactions on Instrumentation and Measurement* (2026) — a biomedical fragmented attention spectral transformer for infrared spectra analysis with applications in biomedical diagnostics.
- Built real-time deepfake detection systems for banking KYC video calls, achieving **73.8% F1** on a private dataset; engineered multimodal ID-card verification combining CV and NLP for document forgery detection.
- Developed a spatial-attention-based behavioral biometric authentication system from raw IMU data (accelerometer, gyroscope, magnetometer), achieving **88% accuracy** with FAR: 0.04 and FRR: 0.08.
- Contributed to CVPR/ICCV challenge-facing forensic systems achieving top-10 finishes across four international competitions in deepfake detection, robust AI-image detection, and low-light enhancement.

AWARDS, CERTIFICATIONS & ACADEMIC ACTIVITIES

- **Awards:** GATE DA Qualified (Graduate Aptitude Test in Engineering in Data Science and Artificial Intelligence).
- **Certifications:** Azure AI Fundamentals; Diploma in Data Science (IIT Madras); Natural Language Processing (NPTEL).
- **Academic Presentation:** Presented at the Research Graduate Meeting, IIT Mandi, 12–13 March 2024.